Notes: Budish, Cramton, and Shim (QJE, 2015)

The High-Frequency Trading Arms Race: Frequent Batch Auctions as a Market Design Response

Contents

1 Big picture

- **Question.** Is the modern high-frequency trading arms race a healthy form of competition that improves markets, or is it a symptom of a flawed exchange design?
- **Paper's answer.** The arms race is mainly a *market-design problem.* Under the continuous limit order book (CLOB), tiny speed advantages let traders capture mechanical arbitrage rents created by serial processing. Those rents induce socially wasteful investment in speed.
- **Core proposal.** Replace continuous-time serial processing with *frequent batch auctions* (FBAs): divide the trading day into very short discrete intervals (for example, 100 milliseconds), treat all orders within an interval as simultaneous, and clear them in a uniform-price double auction.
- **Three-part argument.**
  - *Empirics:* using direct-feed data for the E-mini S&P 500 futures contract (ES) and the SPDR S&P 500 ETF (SPY), the paper shows that correlations break down at millisecond horizons, creating obvious cross-market arbitrage opportunities.
  - *Theory:* in a CLOB, even *symmetrically observed public information* generates stale-quote sniping rents, because order processing is serial rather than batched.
  - *Design:* FBAs eliminate these rents by making time discrete and turning competition on speed into competition on price.
- **Main empirical facts.**
  - In 2011, the median ES–SPY return correlation is only **0.1016** at a 10-millisecond horizon and **0.0080** at a 1-millisecond horizon.
  - The median duration of ES–SPY arbitrage opportunities falls from **97 ms** in 2005 to **7 ms** in 2011.
  - But the per-opportunity profit stays roughly constant at about **0.08 index points** per unit traded.
  - Scaled to all exchanges, the ES–SPY race is worth about **$75 million per year** on average.
- **Main theoretical results.**
  - Continuous trading creates positive spreads and shallow depth *even with no asymmetric information, no inventory costs, and a perfect public signal about fundamental value.*
  - The total sniping rents equal both the cost investors pay via the bid-ask spread and, with endogenous entry, the total amount spent on speed.
  - Frequent batch auctions eliminate sniping in the benchmark model, collapse the spread to zero, deepen the book, and stop the speed race when the batch interval is long enough relative to the speed advantage at stake.
- **Main policy point.** The paper is not mainly saying "HFT is bad." It is saying that *the current rules reward speed in a socially wasteful way.* Change the rules, and the useful parts of electronic liquidity provision can remain while the arms race disappears.
- **Big takeaway.** The key object is not "technology" by itself, but *technology interacting with market design.* Under continuous serial processing, speed is privately valuable because the rules create stale-quote rents. Under discrete-time batch processing, the same technology would mainly compete on price instead.

2 Data and measurement (key objects)

2.1 Institutional environment: the continuous limit order book

- A **limit order** specifies price, quantity, and side (buy or sell).
- The **limit order book** is the set of all outstanding orders; the best buy quote is the bid, the best sell quote is the ask.
- If a new order is priced aggressively enough to hit an outstanding opposite-side order, trade occurs immediately.
- Matching follows **price-time priority**: better prices execute first; ties are broken by which order arrived first.
- The economically crucial design feature is **serial processing**. Orders and cancellations are processed one at a time in order of receipt.

Why this matters. The paper's whole argument is that serial processing, combined with near-simultaneous reactions to public information, mechanically creates races to pick off stale quotes.

2.2 Direct-feed data and why it matters

- The paper uses **direct-feed** data from the Chicago Mercantile Exchange (CME) and the New York Stock Exchange (NYSE).
- CME source: *CME Globex DataMine Market Depth*, covering the E-mini S&P 500 futures contract (`ES`).
- NYSE source: *TAQ NYSE ArcaBook*, covering the SPDR S&P 500 ETF (`SPY`); the paper focuses on SPY within the equity feed.
- Sample period: **January 1, 2005 to December 31, 2011**, with a three-month SPY data gap in 2007 and several calendar-day exclusions (roll dates near ES expiration, half days, corrupted files).
- Final sample: **1,560 trading days**.

Why direct feeds instead of consolidated data?

- Direct feeds record order-book events message by message with exchange-assigned timestamps.
- These are the same data that low-latency trading firms use in real time.
- Consolidated feeds are slower and noisier for the paper's purpose because they aggregate across venues and add processing delays.

2.3 The correlation object

The first main empirical object is the return correlation between ES and SPY as a function of the horizon Δ:

$$\rho(\Delta) = \mathrm{Corr}\big(r_{t,\Delta}^{ES}, r_{t,\Delta}^{SPY}\big).$$

- Returns are computed from bid-ask midpoints.
- The idea is simple: ES and SPY are two large, very closely related instruments tied to the same underlying S&P 500 index.
- At normal human time scales they should move almost one for one; if the market "works" in continuous time, they should remain highly correlated even at very short horizons.

Main fact.

- In 2011, the median correlation is almost 1 at long horizons, but just **0.1016** at 10 milliseconds and **0.0080** at 1 millisecond.
- Over time the market gets faster in the sense that economically meaningful correlations emerge sooner: the ES–SPY correlation reaches 0.50 at about **2.6 seconds** in 2005 but only **142 milliseconds** in 2011.
- Still, in every year correlations are essentially zero at sufficiently high frequency.

Interpretation. Competition makes prices react faster, but it does not make the market truly continuous. There is always a short horizon over which related prices are temporarily out of sync.

2.4 The mechanical-arbitrage object

The second main empirical object is the set of ES–SPY **mechanical arbitrage opportunities**. Conceptually, the strategy is:

- buy the cheaper instrument at its ask,
- sell the more expensive instrument at its bid,
- and reverse when the temporary dislocation closes.

Important implementation details.

- The strategy pays bid-ask spread costs in *both* markets, so the profitability calculation is conservative.
- An opportunity is only counted if the expected profit exceeds a modest threshold of **0.05 index points**.
- An opportunity is only counted if it lasts at least **4 milliseconds**, the one-way speed-of-light lower bound between New York and Chicago.

What the authors want to measure. Not a sophisticated alpha strategy, but the size of the simple, obvious prize that naturally induces an arms race for speed.

2.5 Key descriptive facts from Table I

- Average number of ES–SPY arbitrage opportunities per day: **801**.
- Average quantity per arbitrage opportunity: **13.83 ES lots**.
- Average profit per opportunity: **0.09 index points**, or about **$98.02**.

- Average total daily profits using only NYSE SPY depth: **$79,000**.
- Scaled up using NYSE's market share in SPY to proxy for all exchanges: about **$306,000 per day**, or roughly **$75 million per year**.
- **88.56%** of opportunities are initiated by ES moving first.
- **99.99%** of identified opportunities are "good arbs," meaning they reverse quickly enough to be consistent with temporary dislocation rather than a permanent repricing.

2.6 What these empirical objects are meant to establish

- The correlation object establishes that the market does *not* really function in continuous time.
- The arbitrage object measures the economic consequences of that breakdown.
- The time-series analysis of duration, profitability, and frequency distinguishes between two stories:
  - "competition is eliminating the inefficiency," versus
  - "competition is only changing how fast one must be to capture the prize."

The paper's conclusion is strongly in favor of the second story.

3 Models and empirical specifications (all equations + interpretation)

3.1 Security, public signal, investors, and trading firms

The paper builds a deliberately clean benchmark.

- A security x trades in a continuous limit order book.
- There is a publicly observable signal y that is **perfectly correlated** with the fundamental value of x.
- The signal evolves as a compound Poisson jump process with arrival rate λ_{jump} and jump-size distribution F_{jump}.
- Let J denote the absolute jump size.

Investors arrive with inelastic demand to buy or sell one unit at rate λ_{invest}. If a buyer arrives at time t and trades at time $t' \geq t$ for price p, her payoff is

$$v + (y_{t'} - p) - f_{\text{delay}}(t' - t),$$

with the analogous expression for a seller.

- v is a large positive constant capturing the investor's desire to complete the trade.
- $f_{\text{delay}}(\cdot)$ captures the cost of waiting.

Trading firms have no intrinsic demand; they want to buy below y and sell above y. Initially, the number of trading firms N is fixed.

Interpretation.

- This is a *best-case environment* for the continuous market.
- There is no asymmetric information, no inventory cost, and no search friction.
- So if the CLOB still produces positive spreads and shallow depth, those costs come from market design itself.

3.2 Continuous CLOB equilibrium with exogenous entry

In equilibrium, one trading firm acts as the **liquidity provider** and the other $N-1$ firms act as **stale-quote snipers**.

The liquidity provider posts bid and ask quotes around the current public signal:

$$\text{bid} = y_t - \frac{s}{2}, \qquad \text{ask} = y_t + \frac{s}{2},$$

where s is the bid-ask spread.

If a sufficiently large jump in y occurs, the liquidity provider tries to cancel the stale quote, while the other firms race to hit it first.

Liquidity-provider profits. The liquidity provider earns spread revenue when investors trade, but loses when a stale quote is sniped:

$$\lambda_{\text{invest}} \cdot \frac{s}{2} - \lambda_{\text{jump}} \mathbb{P}\left(J > \frac{s}{2}\right) \mathbb{E}\left[J - \frac{s}{2} \,\middle|\, J > \frac{s}{2}\right] \frac{N-1}{N}. \tag{1}$$

Sniper profits. Each sniper wins the race with probability $1/N$, so expected sniper profits are

$$\lambda_{\text{jump}} \mathbb{P}\left(J > \frac{s}{2}\right) \mathbb{E}\left[J - \frac{s}{2} \,\middle|\, J > \frac{s}{2}\right] \frac{1}{N}. \tag{2}$$

Equilibrium spread. Indifference between providing liquidity and sniping gives

$$\lambda_{\text{invest}} \cdot \frac{s^*}{2} = \lambda_{\text{jump}} \mathbb{P}\left(J > \frac{s^*}{2}\right) \mathbb{E}\left[J - \frac{s^*}{2} \,\middle|\, J > \frac{s^*}{2}\right]. \tag{3}$$

Interpretation of equation (3).

- The left-hand side is the cost investors pay to trading firms via the spread.
- The right-hand side is the total stale-quote-sniping prize created by continuous serial processing.
- The striking point is that $s^* > 0$ even though the model has *none* of the usual microstructure reasons for positive spreads.

Proposition 1.

- There is exactly one quoted unit at the bid and one at the ask.
- Investors trade immediately.
- Large jumps in public information trigger a cancel-versus-snipe race.
- Trading firms are indifferent between being liquidity providers and snipers.
- Total spread revenue paid by investors equals total sniping rents.

Economic meaning. The CLOB causes symmetrically observed public information to be processed *as if it were private information*. That is the paper's deepest theoretical point.

3.3 Multi-unit demand and market thinness

The paper next allows investors to demand multiple units and studies the k-th level of the book. For the k-th quoted unit, equilibrium satisfies

$$\lambda_{\text{invest}} \left(\sum_{i=k}^{\bar{q}} p_i \right) \frac{s_k}{2} = \lambda_{\text{jump}} \mathbb{P}\left(J > \frac{s_k}{2} \right) \mathbb{E}\left[J - \frac{s_k}{2} \,\middle|\, J > \frac{s_k}{2} \right]. \tag{4}$$

- The benefit of posting more depth rises only with the probability that a large investor arrives.
- The cost of being sniped rises with the full amount of stale depth available.

Proposition 2. Spreads are strictly increasing in depth:

$$s_1^* < s_2^* < \cdots < s_{\bar{q}}^*.$$

Interpretation. Continuous trading makes the market *thin*. Large trades face worse prices not because of inventory or private information in the model, but because deeper quotes are especially exposed to being picked off when public information jumps.

3.4 Endogenous entry and the HFT arms race

Now the paper lets firms choose whether to pay a speed cost c_{speed} to reduce latency from δ_{slow} to δ_{fast}, with speed advantage

$$\delta = \delta_{\text{slow}} - \delta_{\text{fast}} > 0.$$

Let N^* denote the equilibrium number of fast firms. The liquidity provider's zero-profit condition becomes

$$\lambda_{\text{invest}} \cdot \frac{s}{2} - \lambda_{\text{jump}} \mathbb{P}\left(J > \frac{s}{2} \right) \mathbb{E}\left[J - \frac{s}{2} \,\middle|\, J > \frac{s}{2} \right] \frac{N-1}{N} = c_{\text{speed}}. \tag{5}$$

Each sniper's zero-profit condition is

$$\lambda_{\text{jump}} \mathbb{P}\left(J > \frac{s}{2} \right) \mathbb{E}\left[J - \frac{s}{2} \,\middle|\, J > \frac{s}{2} \right] \frac{1}{N} = c_{\text{speed}}. \tag{6}$$

Adding these conditions yields

$$\lambda_{\text{invest}} \cdot \frac{s^*}{2} = N^* c_{\text{speed}}. \tag{7}$$

Interpretation of equation (7).

- Investors' total spread payments equal total speed expenditures.
- The private race for speed dissipates the rents created by the market design.
- Endogenous entry does *not* eliminate the underlying stale-quote problem; it only spends resources competing for it.

Proposition 3. With endogenous entry:

- the same basic trading structure remains;
- only fast firms matter in equilibrium;
- total sniping rents = investor spread payments = total expenditure on speed.

3.5 Why the arms race is a prisoner's dilemma

Proposition 4 shows that if the N^* firms could agree not to invest in speed, investors would face the same spread as before in this stylized benchmark, but firms would save the total cost $N^* c_{\text{speed}}$. Yet each individual firm wants to deviate and become fast.

Interpretation.

- The speed race is not privately irrational.
- It is collectively wasteful.
- The market design creates a prisoner's dilemma among trading firms.

3.6 Comparative statics of the arms-race prize

Proposition 5 shows that the total prize in the speed race is

$$\lambda_{\text{jump}} \mathbb{P}\left(J > \frac{s^*}{2}\right) \mathbb{E}\left[J - \frac{s^*}{2} \,\middle|\, J > \frac{s^*}{2}\right].$$

Its key comparative statics are:

- it rises with the **frequency of jumps** λ_{jump};
- it rises when the jump distribution becomes more volatile;
- it is **invariant** to the cost of speed c_{speed}, the absolute speed of fast firms, and the speed gap itself.

Interpretation.

- Better technology does not make the prize disappear.
- It only changes how fast firms must be to capture it.
- This is exactly what the empirics show: durations collapse, but per-opportunity profits remain roughly stable.

3.7 Frequent batch auctions: definition

Frequent batch auctions change two rules at once:

- **time is discrete** rather than continuous;
- **orders are batched** and cleared in a **uniform-price auction** rather than processed serially.

Let the batch interval be $\tau > 0$.

- All orders submitted during a batch interval are treated as arriving at the same time.
- At the end of the interval, the exchange aggregates supply and demand and clears at a single market-clearing price.
- Orders can carry over across auctions.
- Priority depends on how many batch intervals an order has waited, not on microsecond ordering within an interval.
- Information about the new book state is disclosed after the batch is processed.

Important subtle point. It is the combination of *discrete time* and *batch auction clearing* that matters. Discrete time alone is not enough; if messages were still serially processed at the end of the interval, sniping would survive.

3.8 Why FBAs address the problem

Suppose fast firms observe information δ seconds before slow firms. Under FBAs, only information arrivals in the "critical window" of length δ near the end of the batch interval create any temporary asymmetry. So the share of time during which speed matters is approximately

$$\frac{\delta}{\tau}.$$

Proposition 6.

- In FBAs, a slow liquidity provider is only vulnerable during a fraction δ/τ of the trading day, not all of it.
- A fast liquidity provider is not vulnerable to sniping by other fast traders.
- If multiple fast traders try to exploit the same stale quote in the same batch, they compete the price away in the auction. Competition is on *price*, not on microseconds.

Interpretation.

- Continuous trading magnifies tiny speed differences.
- Batch trading dilutes them by the ratio δ/τ.
- And once multiple fast traders react in the same batch, the rent is dissipated *within the auction*, not by costly speed investment outside it.

3.9 Equilibrium of FBAs with exogenous entry

Proposition 7. For any $\tau > 0$ and any fixed $N \geq 2$, there is an equilibrium in which:

- investors trade in the next batch auction after arrival;
- trading firms offer all needed depth at **zero spread**;
- the book is deeper than under the continuous market.

Why zero spread here?

- The model turned off other reasons for spreads.
- Once sniping is removed, Bertrand competition among liquidity providers drives spreads to zero.

Interpretation. The paper is not claiming that real-world spreads would literally become zero. It is claiming that the *sniping component* of spreads disappears.

3.10 Equilibrium of FBAs with endogenous entry

Let Q denote the maximum depth needed to serve investor demand in a batch. A sufficient condition for no one to invest in speed is

$$\frac{p_{\text{jump}}}{\tau}\mathbb{E}[J']Q < c_{\text{speed}}, \tag{8}$$

where p_{jump} is the probability of a jump in the critical window and J' is the total jump size conditional on such a jump.

Using the approximation $p_{\text{jump}} \approx \delta\lambda_{\text{jump}}$, this becomes roughly

$$\frac{\delta}{\tau}\lambda_{\text{jump}}\mathbb{E}[J]Q < c_{\text{speed}}.$$

Proposition 8. If condition (8) holds, then in equilibrium:

- investors trade in the next batch auction;
- slow firms provide all needed liquidity;
- the bid-ask spread is zero in the benchmark model;
- no firm pays for the fast technology.

Welfare comparison.

- *Benefit:* the speed race disappears, saving $N^* c_{\text{speed}}$.
- *Cost:* investors wait up to τ before execution, creating delay cost.

Calibration takeaway.

- If δ is interpreted as year-to-year improvements at the frontier, a batch interval on the order of **10–100 milliseconds** can be enough to stop the race.
- If δ is interpreted as the gap between frontier HFTs and merely sophisticated but slower traders, the relevant interval is more like **100 milliseconds to 1 second**.

- In a variant where information itself arrives in discrete time, the lower bound can be as small as **1–10 milliseconds**.

3.11 Alternative policy responses in the paper's framework

Tobin tax.

- A transactions tax reduces the incentive to snipe.
- But it also raises investors' trading costs directly.
- The paper argues it is a blunt instrument: cutting arms-race profits by 90% would require a tax on the order of **10 basis points**, roughly 10 times the average SPY spread in the sample.

Tax on speed technology.

- Conceptually better than taxing trading, because it targets the arms race directly.
- But still requires very large magnitudes to matter.

Minimum resting times and message ratios.

- These attack symptoms, not the root cause.
- In the model, minimum resting times can actually *worsen* sniping, because they stop liquidity providers from canceling stale quotes.

Random message delays.

- The paper argues they do not eliminate sniping.
- They only randomize who wins and encourage redundant message traffic.

Asymmetric delays such as IEX-style speed bumps.

- These help with stale-quote sniping.
- But they do not fully solve the race to the top of the book and do not transform all speed competition into price competition the way full batch auctions do.

3.12 Computational advantages of discrete-time trading

The paper also emphasizes several implementation advantages of discrete time:

- **Exchange computation is simpler.** Auctions are computationally easy to clear, and exchanges can allocate specific processing time to them.
- **Algorithms face less pressure to trade robustness for speed.** In continuous time, every microsecond matters; in discrete time, traders have a bounded reaction window.
- **The regulatory paper trail is cleaner.** It is easier to reconstruct what happened when events occur at discrete timestamps rather than in an effectively continuous, cross-venue race.
- **Symmetric information dissemination is easier.** Discrete time respects the finite speed of computers and communications.

Interpretation. Even beyond sniping and spreads, discrete time is appealing because it is better aligned with how real computers and networks actually work.

4 Identification and instruments (what makes the estimates credible)

4.1 This is not an IV paper – credibility comes from measurement plus mechanism

The paper is not identifying a causal treatment effect with an instrument. Its credibility comes from combining:

- unusually appropriate data for the object of interest;
- a simple but sharp theoretical mechanism;
- and a market-design counterfactual that directly targets the mechanism.

Interpretation. The paper is persuasive if both halves line up:

- the data must really show stale-quote opportunities that matter economically;
- the theory must show these opportunities are built into the rules rather than accidental.

4.2 Why the empirical evidence is especially credible

- The paper uses **direct-feed** data, not slower consolidated feeds.
- It studies a particularly clean near-arbitrage pair, ES and SPY, where deviations are easy to interpret.
- It reconstructs the book message by message rather than using coarse quote snapshots.
- It explicitly accounts for bid-ask spread costs and the physical speed-of-light lower bound between Chicago and New York.

Interpretation. These choices make the data especially appropriate for studying whether markets "work" at millisecond horizons.

4.3 Why the arbitrage-profit estimates are conservative

The paper itself emphasizes that the $75 million annual estimate for ES–SPY likely understates the true prize.

- The trading rule is intentionally simple.
- It pays spread costs in both markets.
- It only uses NYSE depth for SPY and then scales up conservatively using market-share information.
- It drops opportunities shorter than 4 milliseconds even if they appear in the raw data.

Interpretation. The estimated prize is best read as a lower bound on the value of the race in this one instrument pair.

4.4 Why the theory is especially powerful

The model stacks the deck against the paper's own thesis.

- There is a perfect public signal about value.
- There is no private information.
- There are no inventory costs.
- There are no search frictions.

Why this matters.

- If positive spreads and thin depth still arise, the source cannot be standard adverse selection or inventory risk.
- The only remaining culprit is the continuous serial-processing rule itself.

This is why the paper can claim that sniping is "built in" to the market design.

4.5 Why the "constant of the market design" interpretation is convincing

The paper's time-series evidence is very carefully chosen:

- **duration** of arbitrage opportunities falls a lot;
- **profit per opportunity** stays roughly stable;
- **frequency** mostly tracks volatility.

Interpretation.

- If competition were solving the problem, profitability itself should shrink toward zero.
- Instead, the data say that competition just keeps raising the speed bar.
- Proposition 5 delivers exactly this pattern in the model.

4.6 What is strongest about the FBA proposal

- It targets the mechanism directly.
- It does not rely on punishing traders after the fact or taxing them enough to overwhelm the incentive to race.
- It preserves competition, but redirects it from speed to price.

Interpretation. This is why the design response is much more elegant than the alternative policies discussed in Section VIII.

4.7 Main limitations

- The model is highly stylized and intentionally ignores many real-world features, such as inventory management, asymmetric information, multi-leg hedging, and strategic order splitting.
- Welfare is not fully pinned down because investor delay costs are introduced in reduced-form fashion.
- The empirical evidence focuses on one especially clean cross-market race, ES–SPY, even though the paper strongly suspects the broader race is much larger.
- Implementing frequent batch auctions market-wide would raise practical coordination questions across instruments and venues.

4.8 Relation to the literature

- Relative to the older batch-auction literature, the paper's argument is new: it is not mainly about call auctions versus continuous trading in the traditional sense, but about HFT-era speed races created by serial processing.
- Relative to HFT and microstructure work, the paper isolates stale-quote sniping from other controversial aspects of HFT.
- Relative to market design, the paper is a strong example of the idea that bad outcomes can be traced to rules rather than trader types.

5 Tables and figures

This is a figure-heavy paper with one especially important table. The cleanest way to read the evidence is as a progression from *visual diagnosis* to *economic magnitude* to *theoretical mechanism*.

5.1 Figure I: the market looks continuous at human horizons, but not at machine horizons

Read:

- Panel A (full day), Panel B (hour), and Panel C (minute) show ES and SPY moving almost together.
- Panel D zooms into 250 milliseconds and shows the apparent comovement breaking down.
- After one market jumps and the other has not yet reacted, the arbitrage direction is mechanically obvious.

Why it matters.

- This is the paper's visual thesis in one figure.
- It shows that "continuous trading" is an illusion once the clock runs at the speed relevant for modern trading systems.

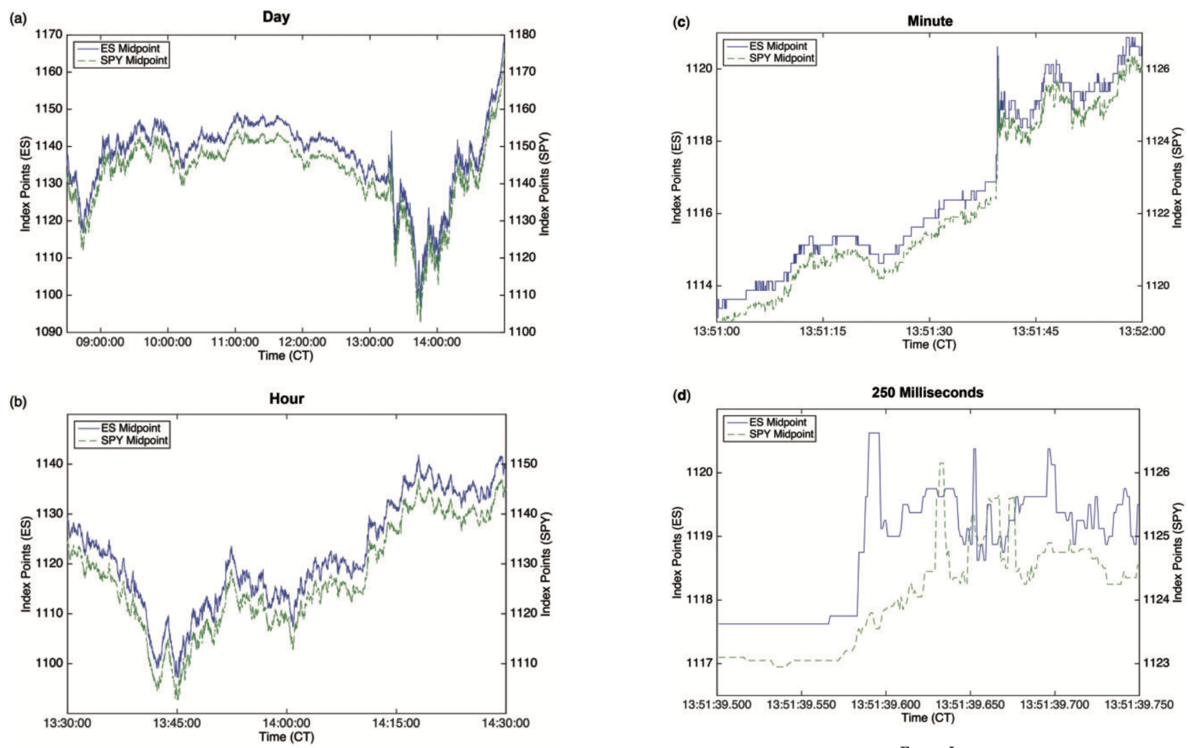

FIGURE I

ES and SPY Time Series at Human-Scale and High-Frequency Time Horizons

FIGURE I

Continued

5.2 Figure II and Figure III: correlation breakdown and how it evolves over time

Read:

- Figure II shows the ES–SPY return correlation as a function of the return interval in 2011.
- The main numbers to remember are **0.1016** at 10 milliseconds and **0.0080** at 1 millisecond.
- Figure III repeats the exercise year by year from 2005 to 2011.
- Correlation recovers faster over time, but never survives at sufficiently high frequency.

Why it matters.

- Figure II documents the existence of the problem.
- Figure III shows that technological progress changes the *timescale* of the problem, not its fundamental nature.

5.3 Table I: how large is the ES–SPY prize?

Read:

- The mean day has **801** arbitrage opportunities.
- The mean arbitrage has quantity **13.83 ES lots** and profit about **$98**.

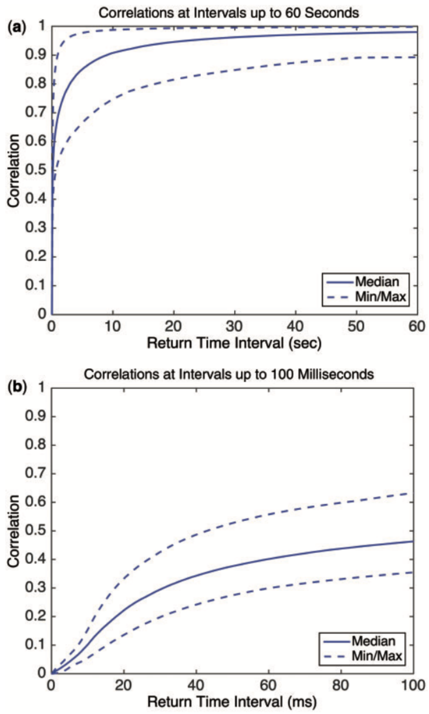

FIGURE II

ES and SPY Correlation by Return Interval: 2011

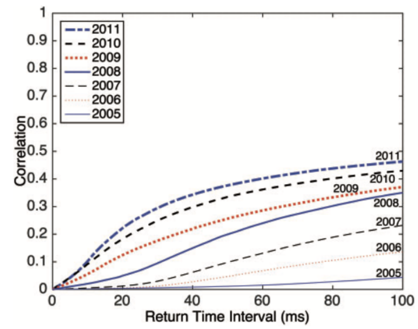

FIGURE III

ES and SPY Correlation Breakdown over Time: 2005–2011

This figure depicts the correlation between the return of the E-mini S&P 500 future (ES) and the SPDR S&P 500 ETF (SPY) bid-ask midpoints as a function of the return time interval for every year from 2005 to 2011. Each line depicts the median correlation over all trading days in a particular year, taken over each return time interval from 1 to 100 milliseconds. For 2005–2008 the CME data is only at 10 milliseconds resolution, so we compute the median correlation for each multiple of 10 milliseconds and then fit a cubic spline. For more details on the data, refer to Section IV.

- Mean total daily profits are about **$79k** using only NYSE SPY depth, and about **$306k** after scaling to all exchanges.
- This implies an annual prize around **$75 million**.

Why it matters.

- The paper is not merely showing a technical irregularity.
- It is showing a prize large enough to rationalize costly real-world investments in fiber, microwave networks, colocation, and specialized hardware.

5.4 Figure IV, Figure V, and Figure VI: duration shrinks, profits persist, frequency follows volatility

Read:

- Figure IV: median arbitrage duration falls from **97 ms** in 2005 to **7 ms** in 2011.
- Figure V: profitability per arbitrage opportunity stays roughly constant, around **0.08 index points**, except for a bump in 2008.
- Figure VI: the number of arbitrage opportunities per day varies with market volatility; the paper reports an R^2 of about **0.87** when daily arbitrage counts are regressed on "distance traveled" by ES prices.

Why it matters.

TABLE I

ES-SPY Arbitrage Summary Statistics, 2005–2011

		Percentile						
	Mean	1	5	25	50	75	95	99
# of arbs/day	801	118	173	285	439	876	2498	5353
Per arb quantity (ES lots)	13.83	0.20	0.20	1.25	4.20	11.99	52.00	145.00
Per arb profits (index pts)	0.09	0.05	0.05	0.06	0.08	0.11	0.15	0.22
Per arb profits ($)	98.02	0.59	1.08	5.34	17.05	50.37	258.07	927.07
Total daily profits, NYSE data ($)	79 k	5 k	9 k	18 k	33 k	57 k	204 k	554 k
Total daily profits, all exchanges ($)	306 k	27 k	39 k	75 k	128 k	218 k	756 k	2,333 k
% ES initiated	88.56							
% good arbs	99.99							
% buy vs. sell	49.77							

Notes. This table shows the mean and various percentiles of arbitrage variables from the mechanical trading strategy between the E-mini S&P 500 future (ES) and the SPDR S&P 500 ETF (SPY) described in Section V.B and Online Appendix A.2.1. The data, described in Section IV, cover January 2005 to December 2011. Variables are described in the text of Section V.B.

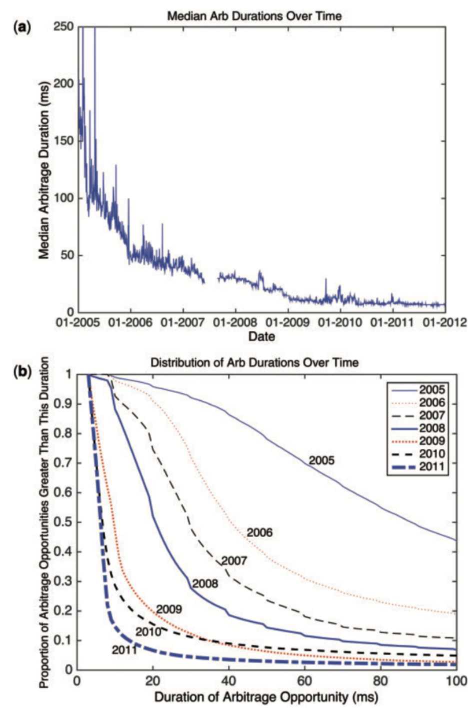

Figure IV

Duration of ES & SPY Arbitrage Opportunities over Time: 2005–2011

Panel A shows the median duration of ES-SPY arbitrage opportunities for each day in our data. Panel B plots arbitrage duration against the proportion of opportunities lasting at least that duration, for each year in our data. We drop opportunities that last fewer than 4 milliseconds, the speed-of-light travel time between New York and Chicago. Prior to November 24, 2008, we drop opportunities that last fewer than 9 milliseconds, the maximum combined effect of the speed-of-light travel time and the rounding of CME data to centiseconds. See Section V.B for details regarding the arbitrage. See Section IV for details regarding the data.

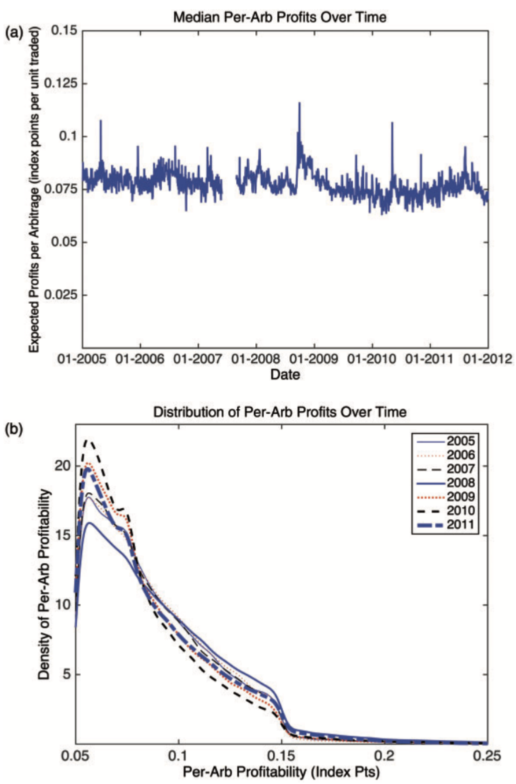

FIGURE V

Profitability of ES & SPY Arbitrage Opportunities over Time: 2005–2011

Panel A shows the median profitability of ES-SPY arbitrage opportunities, per unit traded, for each day in our data. Panel B plots the kernel density of the profitability of arbitrage opportunities, per unit traded, for each year in our data. See Section V.B for details regarding the ES-SPY arbitrage. See Section IV for details regarding the data.

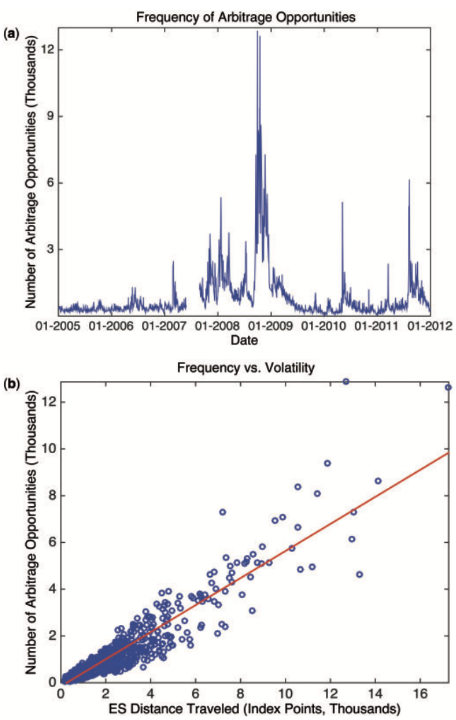

FIGURE VI

Frequency of ES & SPY Arbitrage Opportunities over Time: 2005–2011

Panel A shows the time series of the total number of ES-SPY arbitrage opportunities in each day in our data. Panel B depicts a scatter plot of the total number of arbitrage opportunities in a trading day against ES distance traveled, defined as the sum of the absolute value of changes in the ES mid-point price over the course of the trading day. The solid line represents the fitted values from a linear regression of arbitrage frequency on distance traveled. See Section V.B for details regarding the arbitrage. See Section IV for details regarding the data.

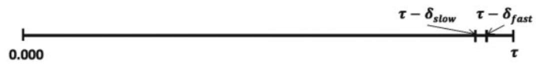

Figure VII

Illustration of How Discrete Time Reduces the Value of Tiny Speed Advantages

- Figure IV says the race has become faster.
- Figure V says the prize has not been competed away.
- Figure VI says the size of the prize is mainly governed by volatility, not by the current frontier of speed technology.

5.5 Figure VII: the core geometry of frequent batch auctions

Read:

- The figure shows a batch interval of length τ, a slow latency δ_{slow}, and a fast latency δ_{fast}.
- Only information arrivals during the narrow critical window of length $\delta = \delta_{\text{slow}} - \delta_{\text{fast}}$ generate any asymmetry between fast and slow traders.
- Hence the fraction of time during which speed matters is δ/τ.

Why it matters.

- Figure VII is the key intuition behind the proposal.
- It makes clear why even modest batching can dramatically reduce the value of tiny speed advantages.

6 Short summary

- **Method.** The paper combines direct-feed evidence on ES–SPY at millisecond horizons with a simple continuous-time limit-order-book model and a market-design counterfactual based on frequent batch auctions.
- **Main empirical result.** Cross-market correlations collapse at very short horizons, producing frequent, obvious mechanical arbitrage opportunities. Their duration shrinks over time, but their profitability does not.
- **Main conceptual result.** Under continuous serial processing, even symmetrically observed public information generates stale-quote-sniping rents. Those rents are built into the market design.
- **Main equilibrium result.** In the benchmark model, the continuous market yields positive spreads, shallow depth, and wasteful speed investment; FBAs eliminate sniping, deepen the book, and can stop the arms race entirely.
- **Main policy lesson.** The right response is not to tax or ban HFT piecemeal, but to redesign the exchange so that competition occurs on price instead of microseconds.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that the HFT arms race is not just a story about faster computers chasing ever smaller private gains.
- It establishes that the *continuous limit order book itself* creates those gains by serially processing reactions to public information.
- It also establishes that a concrete alternative market design, frequent batch auctions, directly targets that flaw.

7.2 Economic intuition

- In a continuous market, if public information changes and related prices have not all updated yet, there is a race to hit stale quotes.
- That race creates rents for being slightly faster, so firms keep investing in speed.
- In a batch auction, those near-simultaneous reactions are pooled together, and competition shifts from "who got there first?" to "who offers the best price?"

7.3 Takeaway

This is a market-design paper disguised as an HFT paper. Its enduring contribution is to show that one should not ask only whether fast traders are good or bad. One must ask what *rules* make speed privately valuable in the first place. The paper's answer is that continuous-time serial processing creates stale-quote rents and an arms race with little social value. Once time is discretized and orders are batched, those rents largely disappear, liquidity improves in the benchmark, and the role of technology becomes much less pathological. That insight has shaped the modern debate about speed bumps, discrete-time trading, and how electronic markets should be organized.